UBER DEMAND–SUPPLY GAP ANALYSIS

A Data Analytics Project Report

Project Overview

This project focuses on analyzing Uber ride request data to identify demand–supply gaps, ride completion performance, cancellation trends, driver availability issues, and customer demand patterns. The analysis is performed using Microsoft Excel, Python, and SQL to generate business insights and provide data-driven recommendations for improving operational efficiency and customer satisfaction.

Tools & Technologies

Microsoft Excel | Python (Pandas, Matplotlib, Seaborn) | SQL | Google Colab

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Project Type

Business Analytics & Data Visualization

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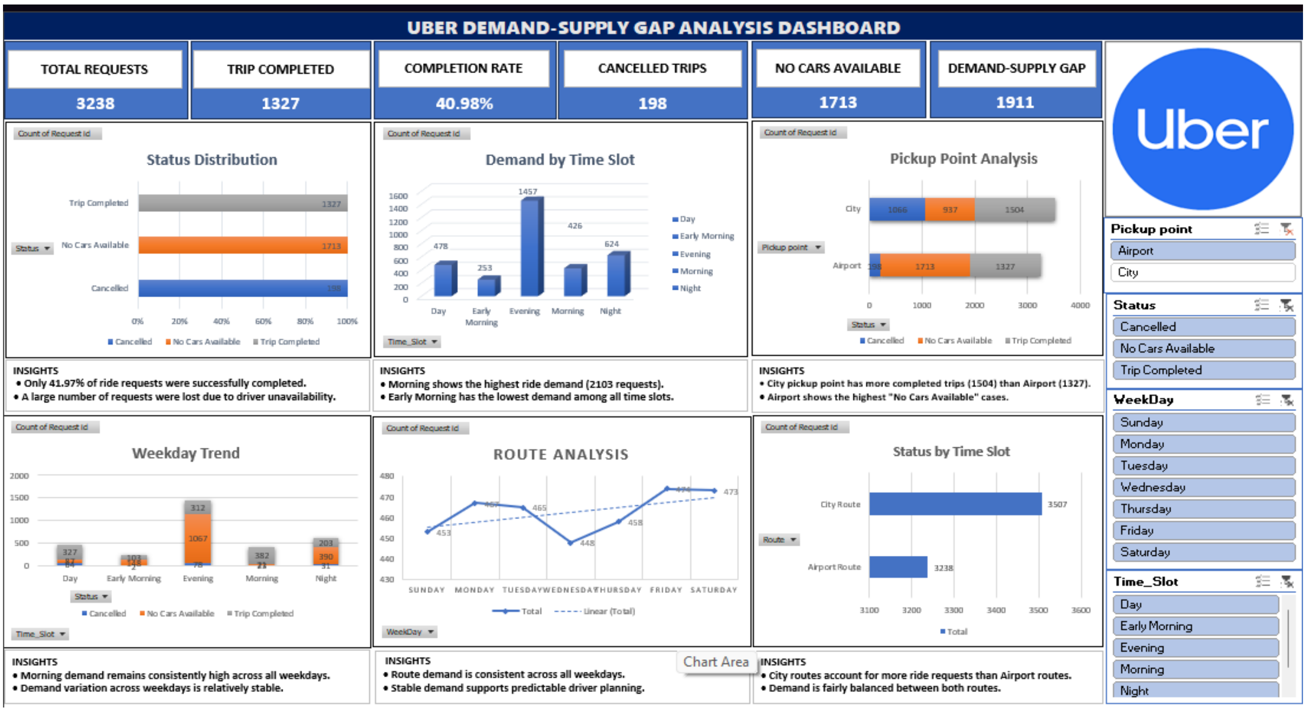
Academic Year

2025–2026

DASHBOARD



1. BUSINESS OBJECTIVES

- Analyze Uber ride request patterns.
- Identify the gap between customer demand and driver supply.
- Evaluate trip completion and cancellation rates.
- Analyze ride demand across different time slots and pickup points.
- Generate actionable business recommendations using data analytics.

2. BUSINESS PROBLEM

Uber is facing operational issues due to driver shortages and ride cancellations.

Evidence from Data

- Total Requests = 6745
- Completed Trips = 2831
- Cancelled Trips = 1264
- No Cars Available = 2650
- Completion Rate = 41.97%

Impact

- Poor customer experience
- Revenue loss
- Lower customer satisfaction

4. Dataset Understanding

Column	Description
Request ID	Ride Request Identifier
Pickup Point	Airport / City
Driver ID	Driver Identifier
Status	Completed / Cancelled / No Cars Available
Request Timestamp	Ride Request Time
Drop Timestamp	Ride Completion Time

5. Data Cleaning & Preprocessing (Python)

Tasks Performed

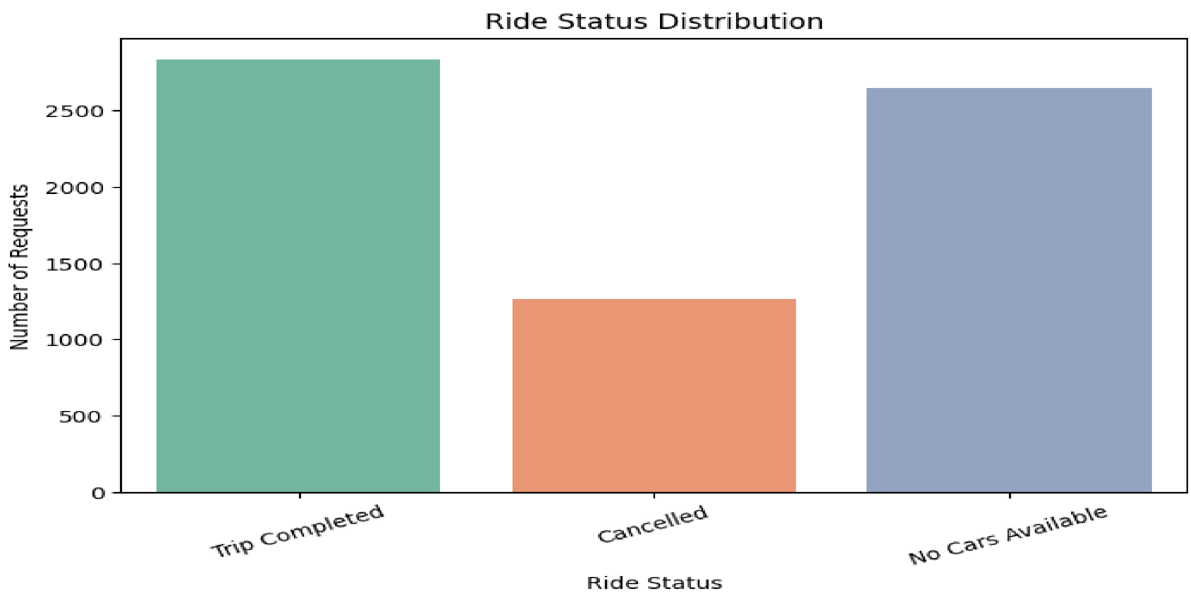
- Missing Value Analysis
- Datetime Conversion
- Request Hour Extraction
- Weekday Extraction
- Time Slot Creation
- Driver Assigned Flag
- Trip Completed Flag

Missing Value Logic

Driver ID and Drop Timestamp contain missing values because some rides were cancelled or no driver was assigned. These are valid business missing values and were retained for analysis.

6. Python Analysis

Chart 1: Ride Status Distribution



Insight:-

- **Only 41.97% rides were successfully completed.**
- **Failed rides (Cancelled + No Cars Available) are more than the completed rides.**

Why:-

- **Driver availability is less based on the customer demand.**
- **During peak hours there is the demand spike.**
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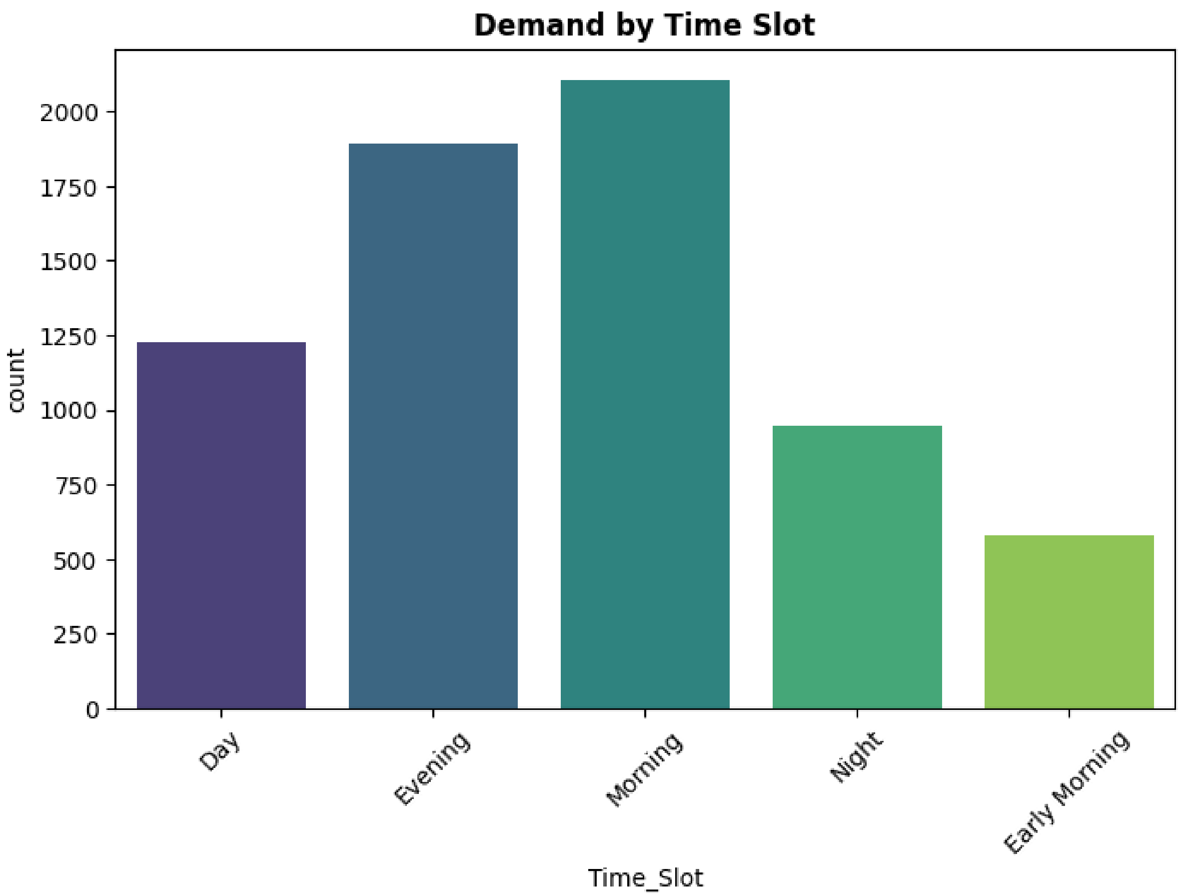
Solution:-

- **Give Incentives to peak hours drivers.**
- **Increase the Driver supply.**

Chart 2: Demand by Time Slot



Insight:-

- **During Morning time slot ,the maximum requests came.**
- **Less demand during the Early Morning.**

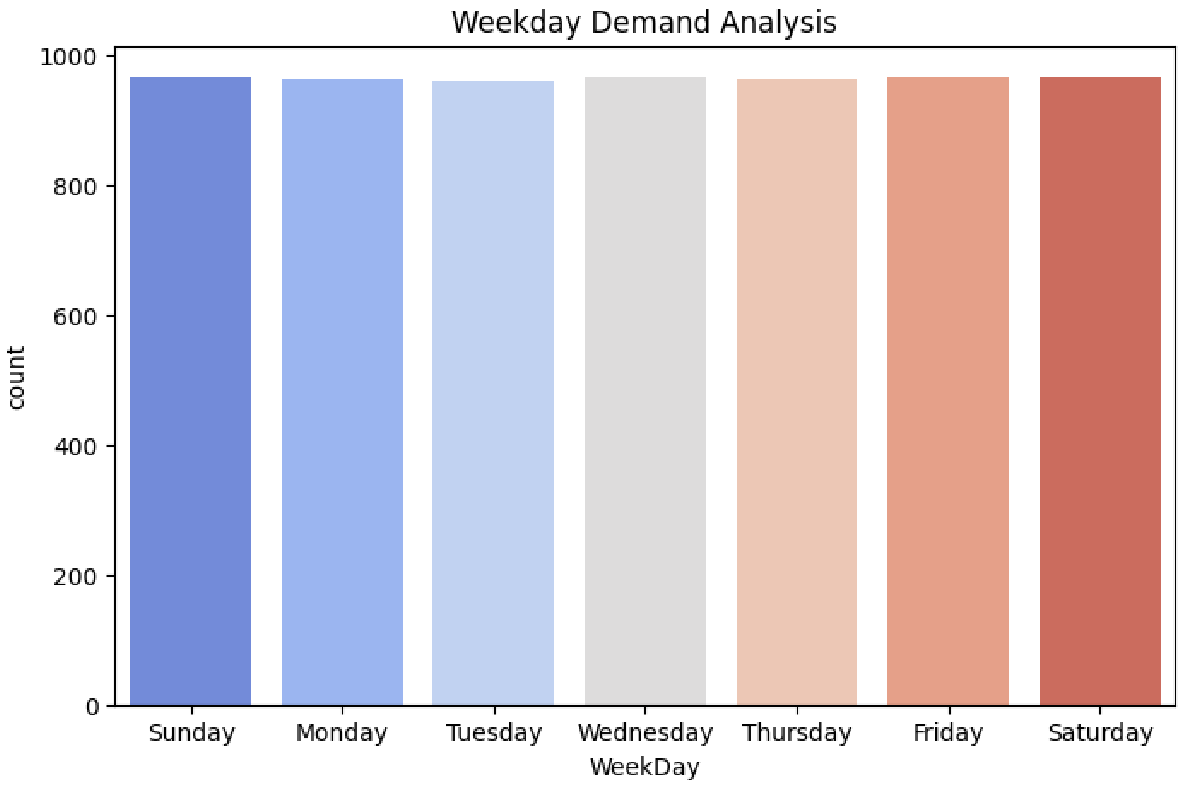
Why:-

- **Due to Office commuting hours the demand increases.**
- **Customers travel more in the morning.**

Solution:-

- **Deploy more drivers during the Morning shifts.**
- **Implement Peak-hour scheduling process.**

Chart 3: Weekday Trend Analysis

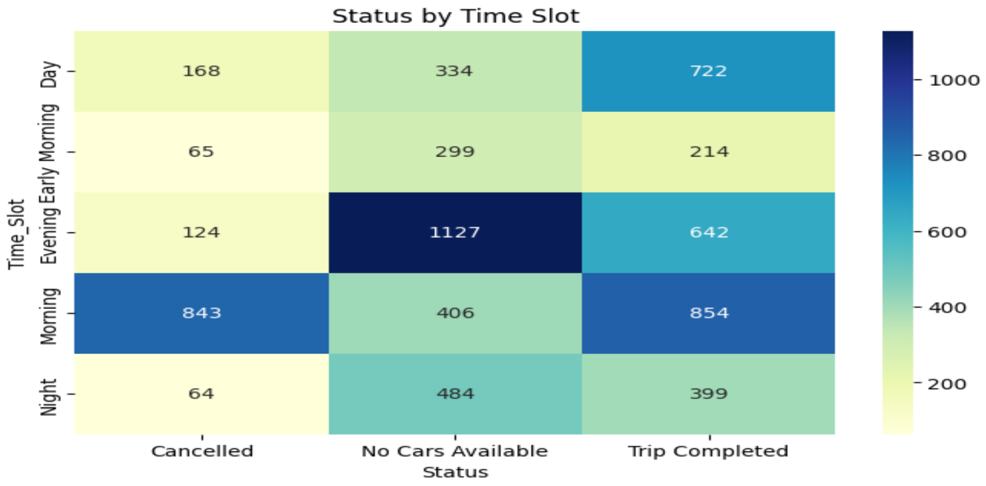


Insight:-
Ride demand remains consistent across all weekdays, indicating stable customer usage patterns with no significant fluctuations.

Why:-
Customers rely on Uber for regular daily commuting and travel, resulting in a balanced distribution of ride requests throughout the week.

Solution:-
Maintain a consistent driver allocation strategy across all weekdays and monitor demand trends to ensure optimal service availability and operational efficiency.

Chart 4: Status vs Time Slot Heatmap:-



Insight:-

- **Morning aur Evening there are maximum rides failures.**
- **No Cars Available cases get increased during peak hours.**

Why:-

- **Demand-supply mismatch.**
- **Available drivers peak demand not get handled.**

Solution:-

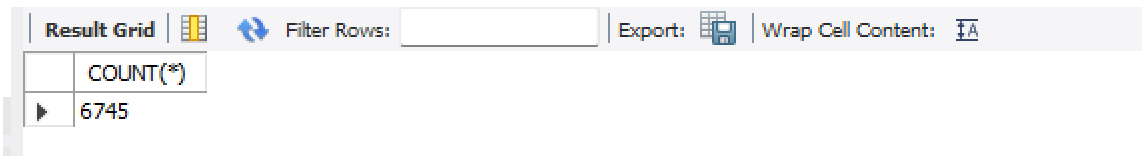
- **Use the Dynamic driver allocation system.**
- **Implement Real-time demand forecasting.**

SQL Analysis-

Query 1: Total Ride Requests

SELECT COUNT(*) FROM uber_analysis;

QUTPUT:



Insight: Dataset contains 6745 ride requests.

Why: Large volume of ride requests indicates significant customer demand.

Solution: Use historical demand trends for forecasting.

Query 2: Weekday Analysis

SELECT WeekDay, COUNT(*)
FROM uber_analysis
GROUP BY WeekDay;

OUTPUT:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

WeekDay	Total_Requests
Friday	965
Sunday	965
Wednesday	965
Saturday	965
Monday	963
Thursday	962
Tuesday	960

Insight:- Demand remains relatively stable across weekdays.

Why:- Daily commuting behaviour drives consistent demand.

Solution:- Maintain balanced driver scheduling throughout the week.

Query 3 Pickup Point Analysis

```
SELECT
Pickup_Point,
ROUND(
SUM(CAST(Success_Flag AS UNSIGNED))*100/COUNT(*),2
) AS Completion_Rate
FROM uber_analysis
GROUP BY Pickup_Point;
```

OUTPUT:

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Pickup_Point	Completion_Rate
Airport	40.98
City	42.89

Insight:-

City contributes more ride demand.

Solution:-

Prioritize driver allocation in city areas.

Query 4 ADVANCED BUSINESS INSIGHT (Demand vs Supply Gap)

```
SELECT
COUNT(*) Total_Requests,
SUM(CASE
WHEN Status='Trip Completed'
THEN 1 ELSE 0 END) Completed,
```

```
SUM(CASE
WHEN Status!='Trip Completed'
THEN 1 ELSE 0 END) Gap
FROM uber_analysis;
```

OUTPUT-

Total_Requests	Completed	Gap
6745	2831	3914

Insight:-

A significant demand-supply gap exists within Uber operations.

More ride requests failed than were successfully completed.

Why:-

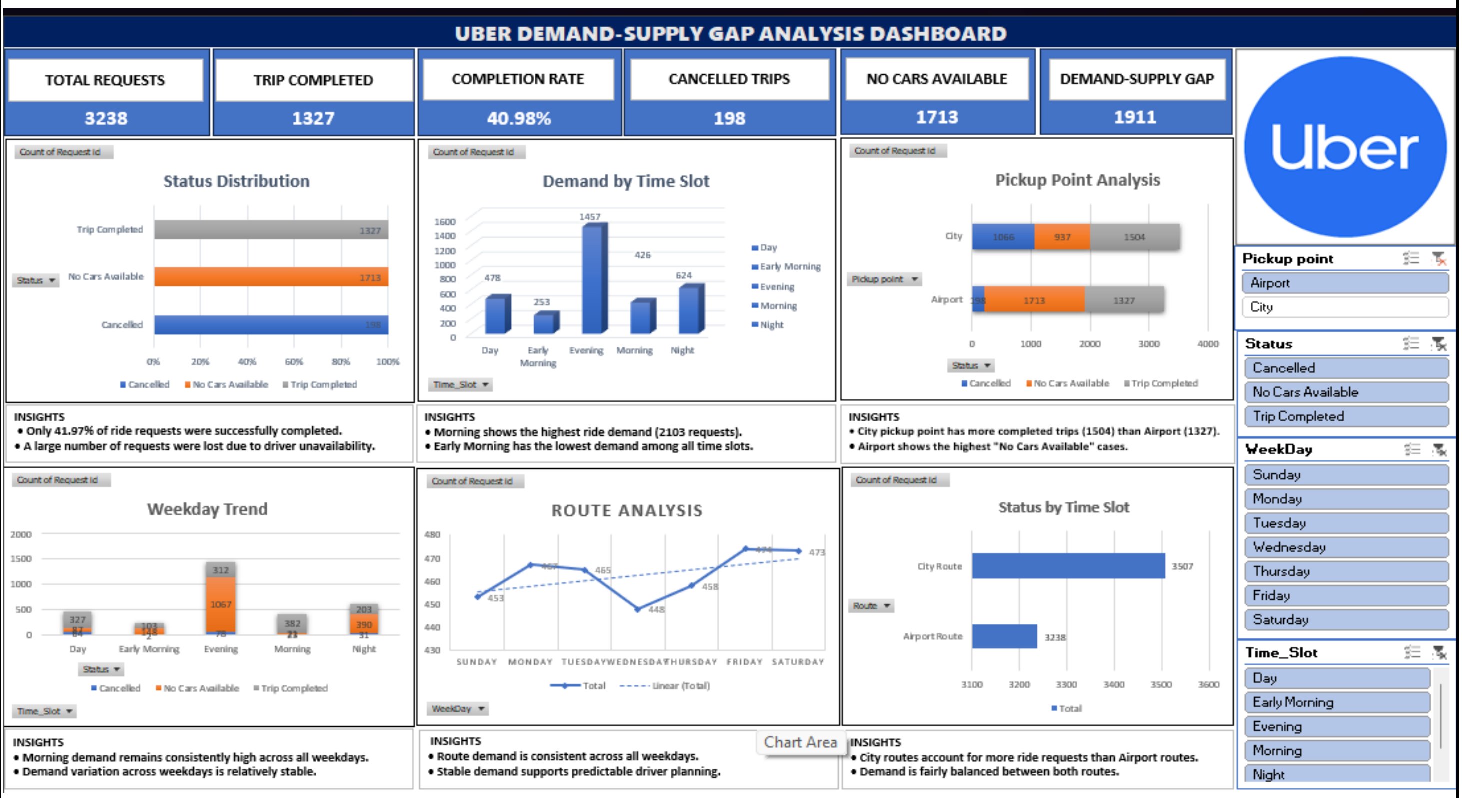
Driver availability is insufficient during peak demand periods.

Solution:-

Increase driver availability, introduce peak-hour incentives,

and implement demand forecasting techniques.

8. Excel Dashboard:-



Key Dashboard Insights:-

1. Demand-Supply Gap = 3914
2. Completion Rate = 41.97%
3. Morning is the busiest time slot
4. City generates highest demand
5. Driver shortage is a major business issue

9. Business Recommendations:-

- Increase driver availability during Morning (5 AM – 10 AM).
- Offer peak-hour incentives.
- Improve airport driver allocation.
- Use demand forecasting models.
- Reduce cancellations through better dispatch management.

10. Conclusion

The analysis identified a significant demand-supply imbalance in Uber operations. Out of 6745 ride requests, only 2831 were successfully completed, resulting in a completion rate of 41.97%. Morning hours and city pickup points generated the highest demand, while driver shortages and cancellations were the primary reasons for ride failures. By increasing driver availability, implementing demand forecasting, and improving driver allocation strategies, Uber can significantly improve customer satisfaction and operational efficiency.